Sampling and Intended Use

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

One current intended-use and coverage record.

Define condition cells, compare counts and propose evidence for a missing condition.

Keep one habit: Fresh sampling W1–W3/W5 fields preserve earlier audit responses; selected W4 remains the outdoor plan.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Label Quality. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Python and Table Readiness. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from The ML Lifecycle. Give an example and a mistake that the idea helps you avoid.

Label Quality

> **My remembered idea, example and correction**

Python and Table Readiness

> **My remembered idea, example and correction**

The ML Lifecycle

> **My remembered idea, example and correction**

W1 · Define intended use

> Current sampling questions have fresh answer fields; earlier audit responses remain preserved.

Complete current sampling reference

Current Coverage v1 intended use has day/evening × cameras A/B. A: day/A9,day/B0,evening/A1,evening/B0;10photos/10plants. B: each cell3;12photos/12plants. Sources invented, species/label counts unknown, no model predictions. Copying each A photo ten times gives100rows/10plants and still no camera-B coverage.

Separate picture:12midday photos/12plants plus12same-plant midday views gives24photos/12plants, no evening evidence.

Current task and response

- State the four intended light/camera combinations and explain why photo count is different from distinct-plant count.

My coverage reasoning and evidence

▼ **Review · Earlier audit task and saved response**

For Robot practice v1, state what one row is intended to represent, the source, the unit and the meaning of None.

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Compare supplied coverage

> Current sampling questions have fresh answer fields; earlier audit responses remain preserved.

Complete current sampling reference

Current Coverage v1 intended use has day/evening × cameras A/B. A: day/A9,day/B0,evening/A1,evening/B0;10photos/10plants. B: each cell3;12photos/12plants. Sources invented, species/label counts unknown, no model predictions. Copying each A photo ten times gives100rows/10plants and still no camera-B coverage.

Separate picture:12midday photos/12plants plus12same-plant midday views gives24photos/12plants, no evening evidence.

Current task and response

- Using Coverage v1 A/B manifests, report day/A, day/B, evening/A and evening/B counts for each sample. Explain why B addresses a known gap without proving higher accuracy. If A photos are each copied ten times, state photo and distinct-plant counts and which cells remain missing.

My coverage reasoning and evidence

▼ **Review · Earlier audit task and saved response**

Explain each number in 5 4 2 1. How many extra ID occurrences are there, and how many rows belong to a repeated-ID group?

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Repair a copied-data claim

> Current sampling questions have fresh answer fields; earlier audit responses remain preserved.

Complete current sampling reference

Current Coverage v1 intended use has day/evening × cameras A/B. A: day/A9,day/B0,evening/A1,evening/B0;10photos/10plants. B: each cell3;12photos/12plants. Sources invented, species/label counts unknown, no model predictions. Copying each A photo ten times gives100rows/10plants and still no camera-B coverage.

Separate picture:12midday photos/12plants plus12same-plant midday views gives24photos/12plants, no evening evidence.

Current task and response

- A teacher says100 copied indoor day photos prove reliable evening camera-B performance. What evidence is missing, and which counts can increase without closing that gap?

My coverage reasoning and evidence

▼ **Review · Earlier audit task and saved response**

A learner changes None to 0 and -2 to 2, then reports a clean dataset. Explain why this is not a justified repair and what record should be kept.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Plan a revealing collection

Current sampling questions have fresh answer fields; earlier audit responses remain preserved.

Complete current sampling reference

Current Coverage v1 intended use has day/evening × cameras A/B. A: day/A9,day/B0,evening/A1,evening/B0;10photos/10plants. B: each cell3;12photos/12plants. Sources invented, species/label counts unknown, no model predictions. Copying each A photo ten times gives100rows/10plants and still no camera-B coverage.

Separate picture:12midday photos/12plants plus12same-plant midday views gives24photos/12plants, no evening evidence.

Current task and response

- Propose collection that directly addresses one zero cell in A. State what IDs, labels and conditions to retain, which uncertainty remains and how a fresh final evaluation differs from development collection.

My coverage reasoning and evidence

▼ **Review · Earlier audit task and saved response**

Someone duplicates each row ten times and claims the export now covers more kinds of trips. Explain what changed and what did not.

My reasoning and evidence

Try alone · Selected experiment task

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- This selected task asks for a collection/evaluation plan, not an invented score. The source gives no outdoor counts, camera specifications, labels or predictions. Name missing conditions, preserve shared plant IDs when collecting related views, distinguish development collection from fresh final evaluation and state what remains unknown. No particular sample size guarantees adequate accuracy.

- **Training** plant photos are all indoors on one phone. **Intended use** includes outdoor photos on other cameras. Plan a sampling check before making a general accuracy claim.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Changed independent case

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 9

Project · Keep the experiment evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Freeze the Plant Classifier intended-use statement and a coverage table for species, lighting, camera, and health class.

- The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	

Evidence	My record
One remaining question	

MAIN LESSON · SECTION 10

Finish · Judge your evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- - I can explain the mechanism.
  - I can show evidence.
  - I can handle a changed case.
  - I can name a limit.

  Mark each as not yet, with help or independently, and attach supporting work.

- Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Earlier trip-audit completion record: retain the saved response as review. Use the current W2 condition-coverage task for this week's supported practice.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- Use the worked case for Dataset inspection and cards. Proposed explanation: “A complete card certifies data quality”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier learning target

REFERENCE · SECTION 5

Reference · Earlier learning target

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- **Training** uses indoor photos from one phone; deployment uses outdoor photos from several phones. Specify two evaluation slices and explain why adding more indoor photos may not resolve the gap.

▼ **Extra practice · Earlier complete case**

REFERENCE · SECTION 6

Extra practice · Earlier complete case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- **Training** photos use indoor white backgrounds; **intended use** is outdoors.
- Identify the coverage gap.
- Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ Extra practice · Changed case

REFERENCE · SECTION 7

Extra practice · Changed case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- **Training** photographs all show mature plants in bright light, but **intended use**rs photograph seedlings indoors. What claim is unsupported and what data is needed?

- My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 7 Test and reflect

▼ Extra practice · Design a revealing test

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Extra practice · Earlier transfer**

REFERENCE · SECTION 11

Extra practice · Earlier transfer

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- **Training** uses indoor photos from one phone; deployment uses outdoor photos from several phones. Specify two evaluation slices and explain why adding more indoor photos may not resolve the gap.

- Record support used: none, hint or teacher explanation.

My independently reasoned answer

▼ **Reference · Earlier W4 record**

REFERENCE · SECTION 15

Reference · Earlier W4 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Schema for this exercise: one intended row per distinct trip ID; each row is an (ID, duration) pair. Duration is numeric minutes or None for unmeasured. Negative durations are invalid; a recorded zero is allowed. Count extra ID occurrences beyond the first and do not silently alter the raw data.

- A new fictional indoor trip export has (P,4), (Q,None), (Q,None), (R,-1), (S,6), (T,0), with durations in minutes and the same **schema**.

  Count raw rows, distinct IDs, missing durations, negative durations and extra ID occurrences. Identify overlap. Draft a short **dataset card** stating source, row meaning, unit, findings, unchanged version and one excluded use.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 18

Extra practice · Pictured case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- **Sample** for the conditions in which the plant classifier will actually be used.
- **Training** set: 12 windowsill plants photographed at midday. **Intended use** includes evening desk photos; no evening examples have been collected.
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.
- Which collection would address the intended-use gap most directly? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.



Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ **Reference · Project portfolio milestone**

REFERENCE · SECTION 19

Reference · Project portfolio milestone

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Project: Responsible Plant Classifier
- Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.
- For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice · Predict, check and repair**

REFERENCE · SECTION 20

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- **Sample** for the conditions in which the plant classifier will actually be used.
- Which labelled step matches this detail? Collect consented fictional or teacher-created evening examples; keep plant groups separate across development and final evaluation.
- 1. Identify gap
- 2. Declare coverage

- 3. Plan comparison

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ Check the first explanation

After your attempt, compare with the supported explanation: Plan comparison Collect consented fictional or teacher-created evening examples; keep plant groups separate across development and final evaluation.

Read the picture in order. Collect consented fictional or teacher-created evening examples; keep plant groups separate across development and final evaluation.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: Collect 12 more midday windowsill photos of the same plants. Which conclusion is supported by the supplied facts?

- 1. This one example guarantees correct results for every future **input**.

- 2. We can decide the result without examining the changed **input** or the procedure.

- 3. The evening-coverage gap remains.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: The evening-coverage gap remains. More rows in the same condition do not supply **evidence** about the missing condition.

Repair and keep the record

Review Label Quality (ai-4/6). Goal: Write an observable label rule with boundaries and a review option. Short practice: Why should two annotators not vote away Z's missing **evidence**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Python and Table Readiness (ai-4/4). Goal: Specify a table **schema**. Short practice: Why should the **missing value** not silently become zero?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review The ML Lifecycle (ai-4/1). Goal: Distinguish **fitting** a rule from applying a fixed rule. Short practice: Should the team silently retune on the changed test?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
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> | | |
> |---|---|
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> | | |
> | | |
> | | |

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

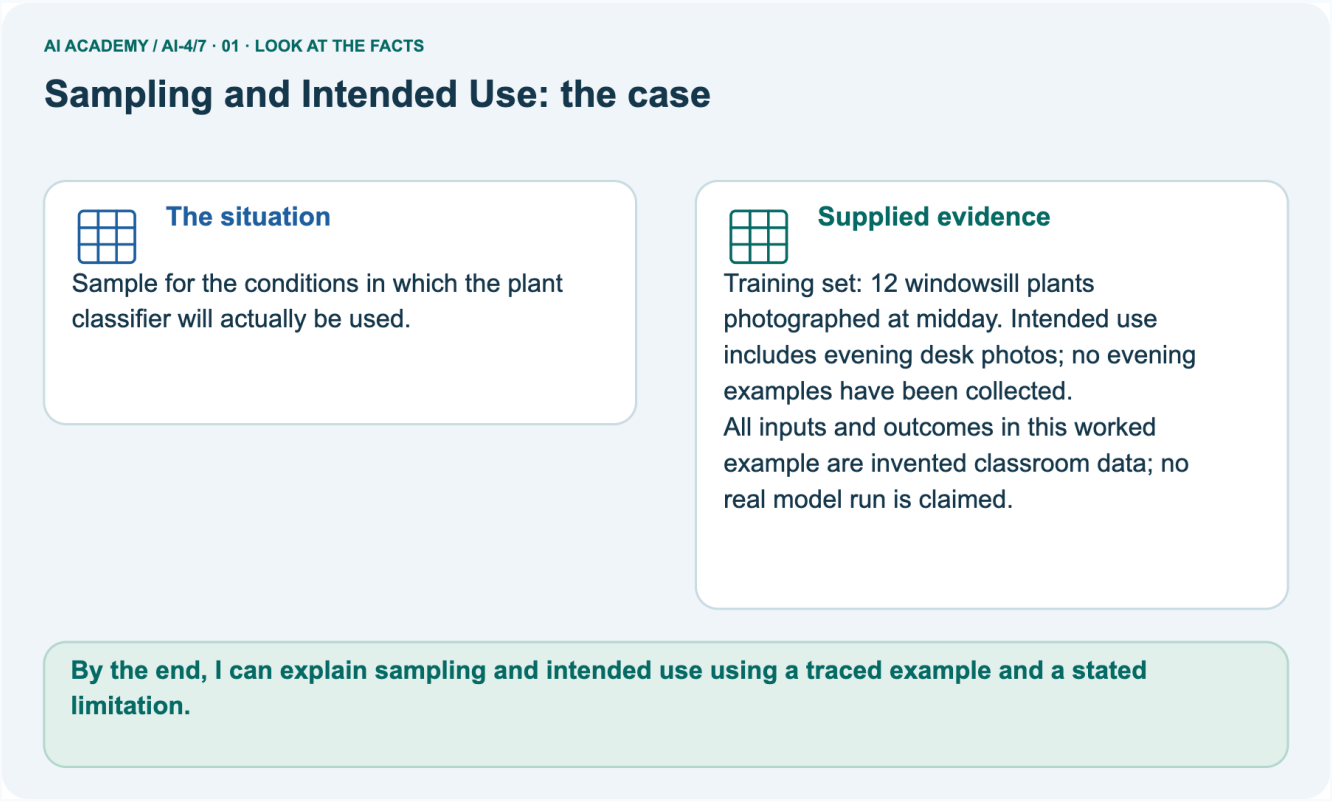
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the relevant picture instead of repeating every task.

01. Sampling and Intended Use: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

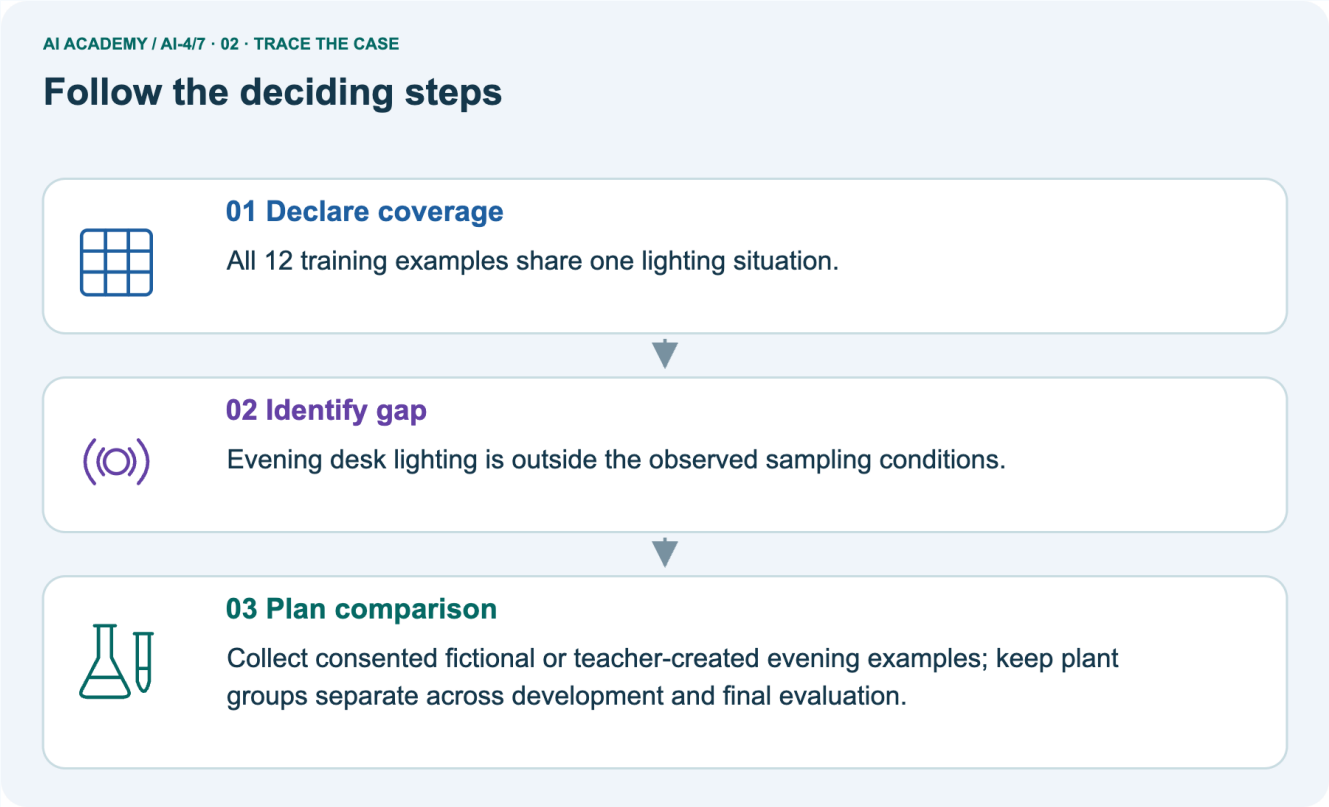
Sample for the conditions in which the plant classifier will actually be used. Supplied facts: **Training** set: 12 windowsill plants photographed at midday.

Intended use includes evening desk photos; no evening examples have been collected. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



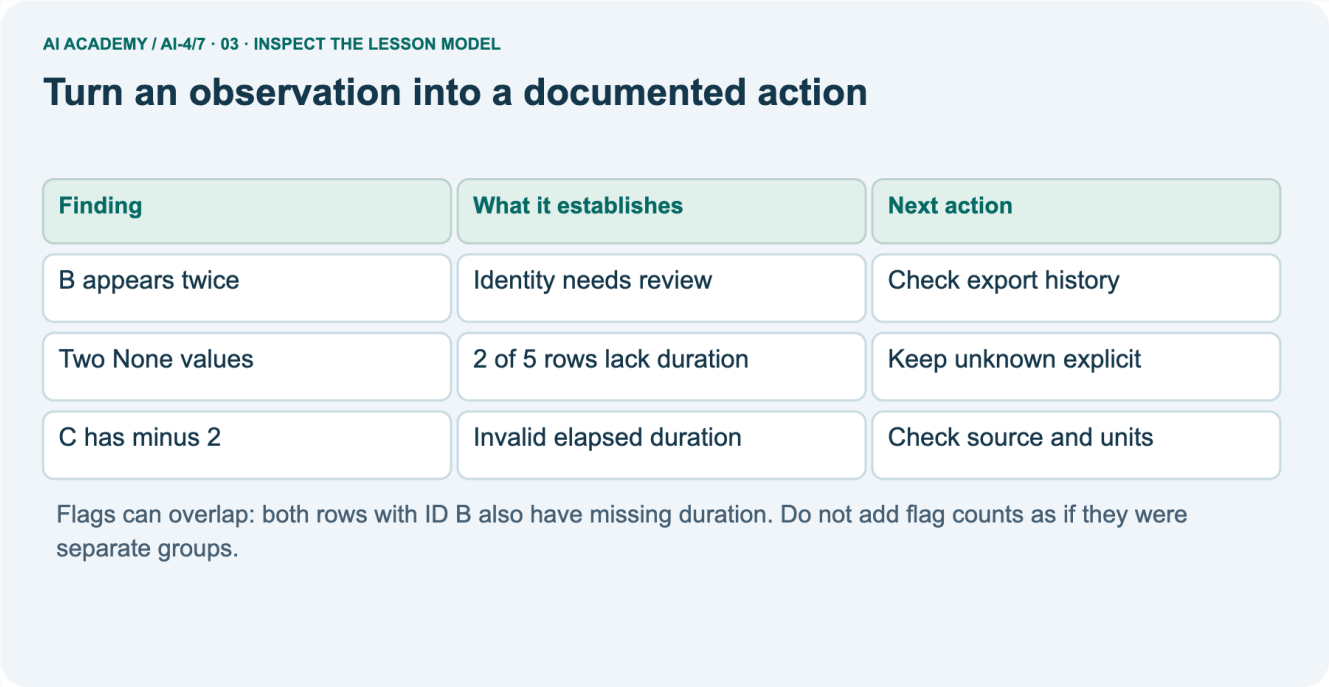
Follow the labelled picture with the matching explanation. Open the vector diagram

Declare coverage: All 12 **training** examples share one lighting situation. → Identify gap: Evening desk lighting is outside the observed sampling conditions. → Plan comparison: Collect consented fictional or teacher-created evening examples; keep plant groups separate across development and final evaluation.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Turn an observation into a documented action



Follow the labelled picture with the matching explanation. Open the vector diagram

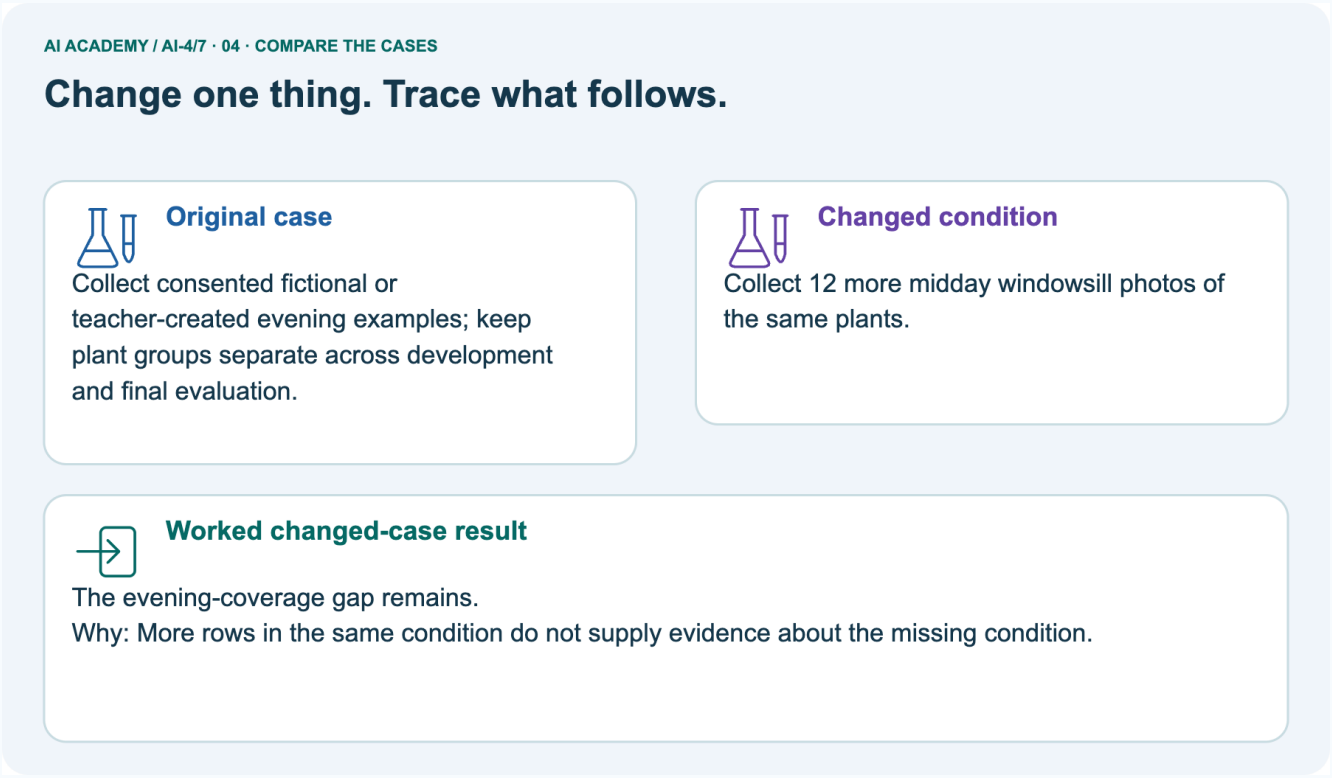
Turn an **observation** into a documented action. B appears twice | Identity needs review | Check export history. Two None values | 2 of 5 rows lack duration | Keep unknown explicit.

C has minus 2 | Invalid elapsed duration | Check source and units

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

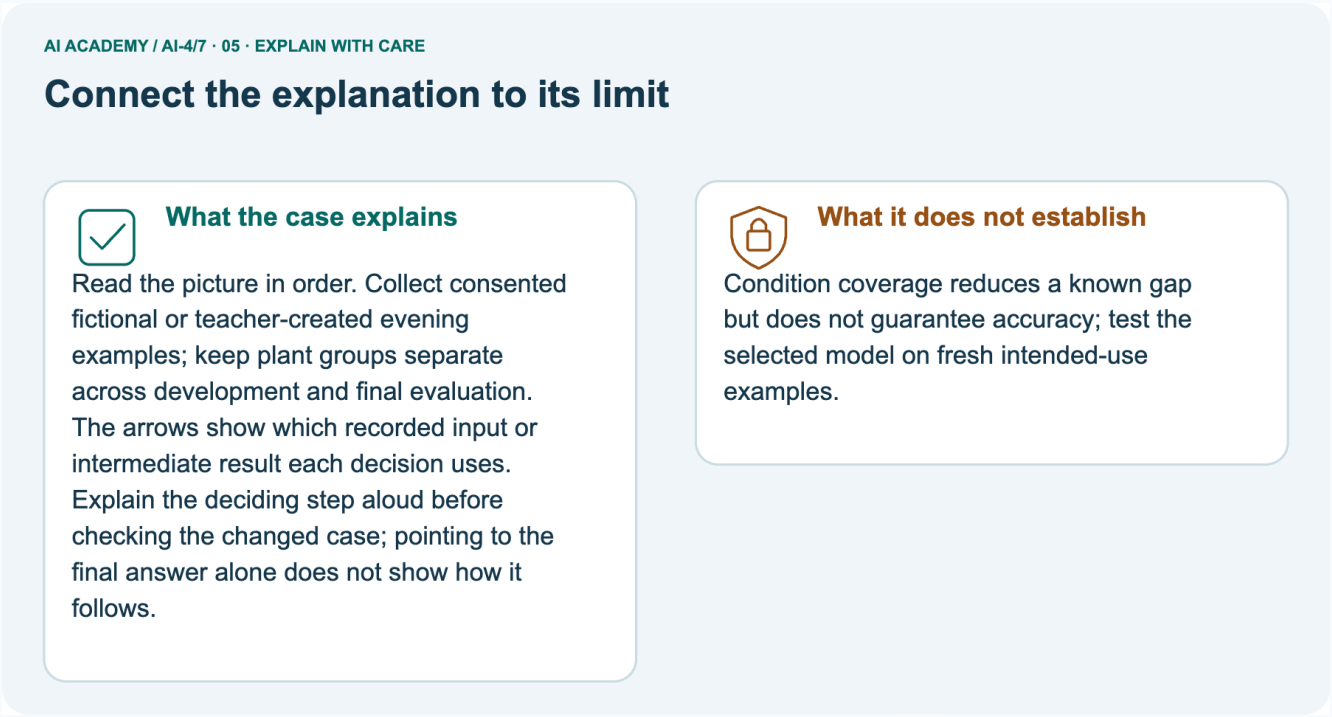
Original case: Collect consented fictional or teacher-created evening examples; keep plant groups separate across development and final evaluation.

Changed condition: Collect 12 more midday windowsill photos of the same plants. Expected result: The evening-coverage gap remains. Reason: More rows in the same condition do not supply **evidence** about the missing condition.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Collect consented fictional or teacher-created evening examples; keep plant groups separate across development and final evaluation.

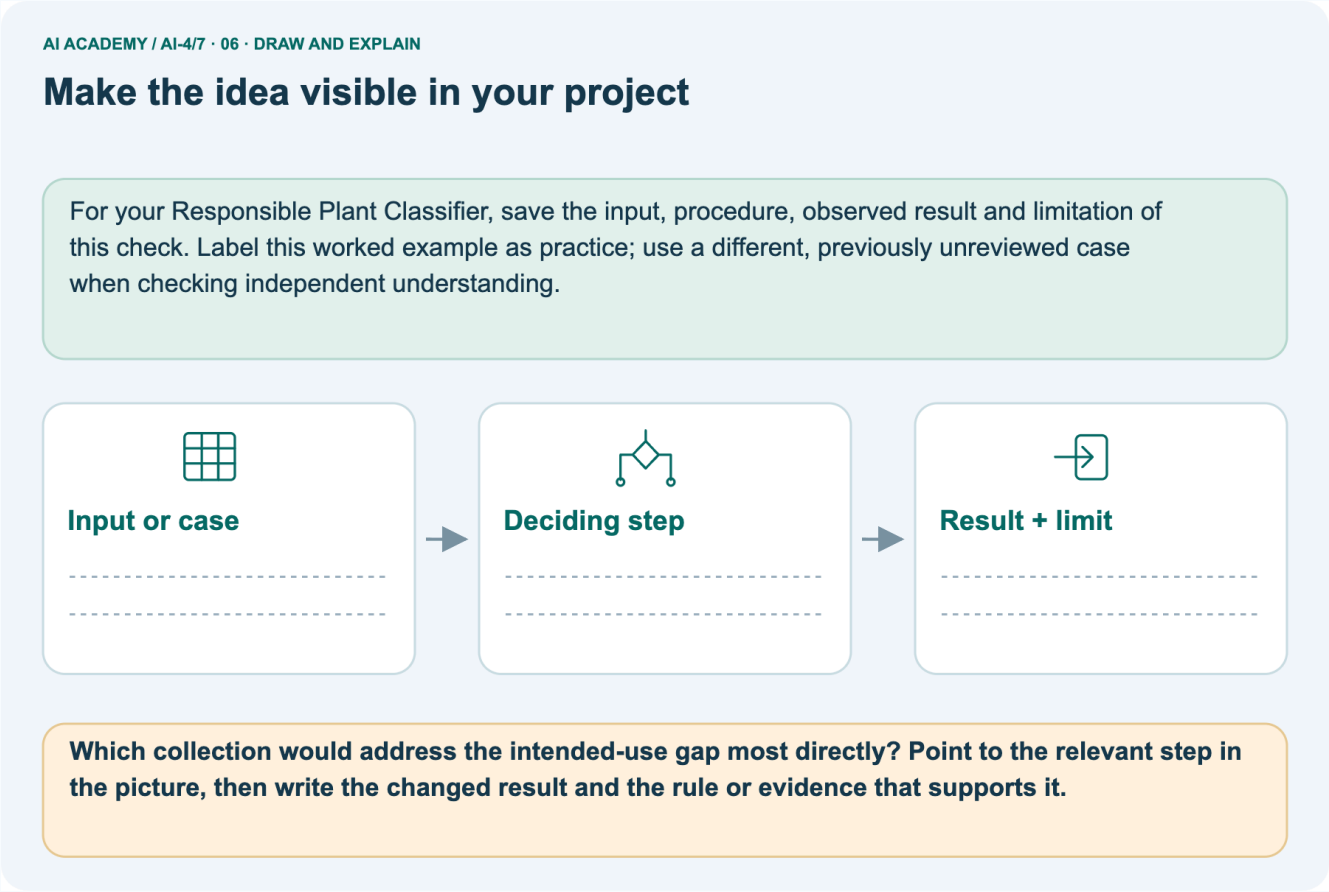
The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Condition coverage reduces a known gap but does not guarantee accuracy; test the selected model on fresh intended-use examples.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which collection would address the intended-use gap most directly? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Sampling and Intended Use: the case

02. Follow the deciding steps

03. Turn an observation into a documented action

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.